UNIVERSITY OF WATERLOO



MTE 544 – Lab 2 Conclusions

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Using:

Station MMEPC157, Robot 06

Section 1

To tune the controller for the goal point case, we followed a standard approach tuning each portion one at a time. To begin, we isolated for the proportional gains K_p in both linear and angular components. We started with a higher $K_p$but it produced an overdamped system with high oscillations. So, we incrementally decreased the proportional gain to get less overshoot. However, there was still minor error overshoot, and the linear components exhibited steady state error. To address the angular overshoot, we introduced the derivative gains. The derivative term acts as predictive damping, counteracting the proportional command as the rate of error change increases. By tuning this gain to 0.2, the controller successfully applied braking as the robot approached its target heading. To eliminate the steady state error from the P controller, we introduced an integral gain of 0.2, so that the controller accumulated small, persistent error over time that would eventually be enough to force the motors to complete the last bit of motion.

Figure 1 displays the linear and angular errors for the P and PID controller getting to a commanded point, along with their values for x, y and theta over time. The linear error responses of the P and PID controllers are nearly identical, with both gradually converging toward zero over the duration of the motion. Both angular controllers also have a very similar response, although the P controller demonstrates slightly faster initial convergence, whereas the PID controller exhibits a slightly smoother angular response.

Both controllers demonstrate similar position tracking, with their X and Y responses following similar trajectories and reaching approximately the same final position. A slightly greater difference is observed in the angular response, where the PID controller exhibits a more gradual initial transition.

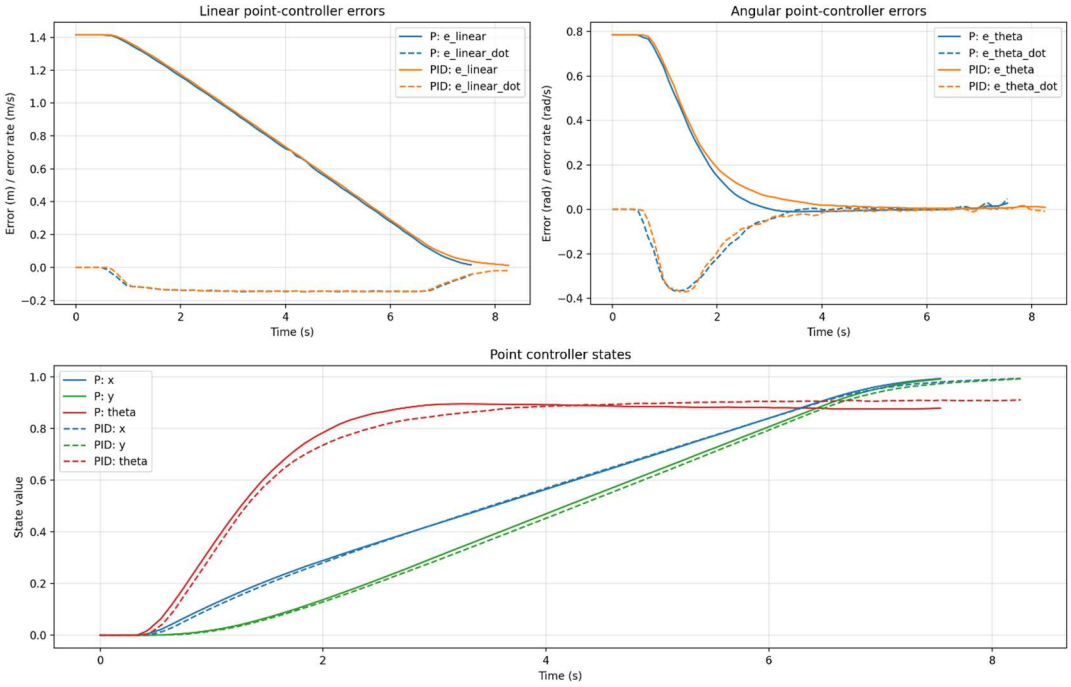


Figure 1: Linear and Angular Errors and State Over Time for P and PID Controller

Figure 2 shows the (x, y) path the robot takes as well as the linear error phase plane and angular error. The XY trajectories of both controllers again are very similar with the proportional controller being slightly sharper or more aggressive overall. The linear error phase planes also exhibit nearly identical behaviour, indicating similar position-error behaviour. Slightly greater differences are observed in the angular phase plane with the PID controller being slightly less sharp as we saw above.

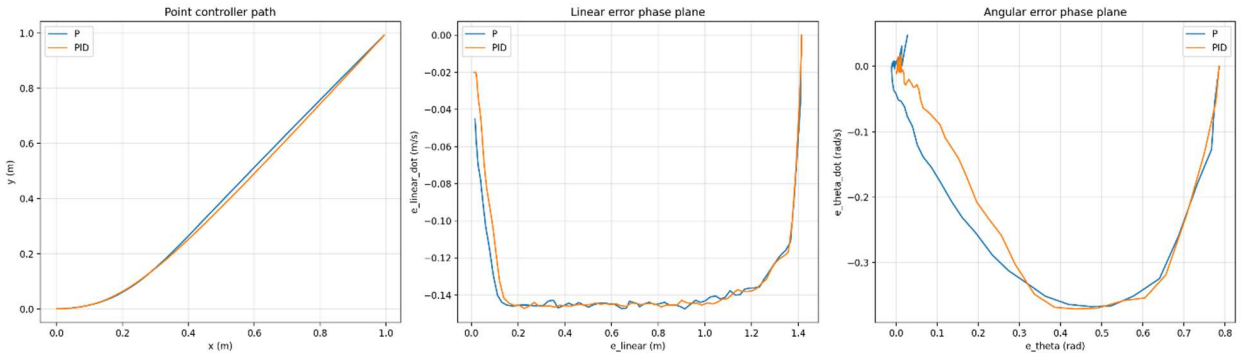


Figure 2: Controller Path and Error Phase Planes

Figure 3 and Figure 4 show the path of the robot relative to its desired path for the parabola and sigmoid trajectory. The robot successfully follows the desired parabolic path, maintaining the overall trajectory shape. There is some overshoot initially, but it decreases as the robot progresses. The trajectories eventually nearly overlap, with only minor steady state error remaining near the end. The robot only somewhat successfully follows the desired trajectory path. Initially, the actual position overcorrects, resulting in some tracking error. However, the overshoot decreases through the middle and end portions of the motion, with only minor deviations remaining.

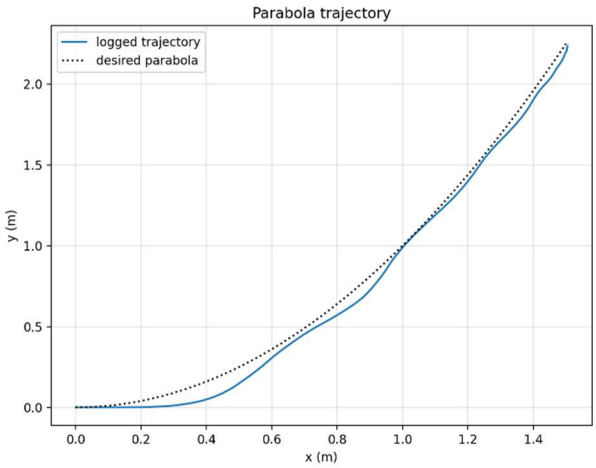


Figure 3: Parabola Trajectory

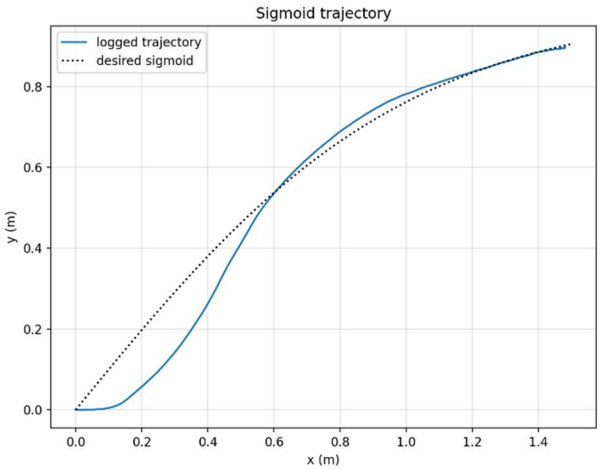


Figure 4: Sigmoid Trajectory

Section 2

Numerical metrics for assessing the P controller vs PID controller were calculated from the graph data and shown below.

Table 1: P vs PID Metrics

Metric	P controller	PID controller
Linear Settling time (agility)	~7.5 s	~7.5 s
Angular Settling time (agility)	~ 3s	~ 4s
Linear Steady-state error (Accuracy)	0.0 m	0.0 m
Angular Steady-state error (Accuracy)	0.0 rad	0.0 rad
Linear Overshoot	0.0 m	0.0 m
Angular Overshoot	0.0 rad	0.0 rad

The numerical comparison highlights the differences in performance between the P and PID controllers. In terms of agility, both controllers achieved the same settling time for linear tracking, and also angular steady state error. Regarding accuracy, both controllers demonstrated no steady-state errors, consistent with the similar error responses observed in Section 1. Finally, both controllers did not have any overshoot as they were both overdamped. Overall, the results suggest that the PID controller preforms slightly better due to the lower angular steady state error.